Lecture 1: Introduction to AI/ML and Surrogate Modeling for Accelerators

Physics Foundations, Forward Neural Digital Twins, and Closed-Loop Orbit Control

Event: AI Applications in Accelerators Tutorial & Mini-Hackathon
Venue: Korea Multi-purpose Accelerator Complex (KOMAC) / KAERI
Speaker: Dr. Chong Shik Park | **Duration:** 80 Minutes (09:10 – 10:30)
Companion Challenge Notebook: `Single_Notebook_Beam_Orbit_Predictor.ipynb`

Abstract

Modern particle accelerator facilities operate with thousands of tightly coupled magnet actuators and diagnostics under sub-micron beam stability tolerances. Classical linear feedback systems face fundamental bottlenecks in non-linear regimes where magnetic core saturation and power supply limits degrade control fidelity. This lecture provides the theoretical and computational foundation bridging transverse beam dynamics, Hill's equation, and the Orbit Response Matrix (ORM) with cutting-edge supervised machine learning and reinforcement learning. We explore the design and training of neural surrogate digital twins to model cubic saturation, compare classical singular value decomposition (SVD) with model-based bounded optimization (`L-BFGS-B`), and investigate physics-informed closed-loop policy agents with adaptive damping for the afternoon Beam Orbit Alignment Mini-Hackathon.

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1 Lecture Timeline & Curriculum Overview

Time	Duration	Core Module Focus
09:10 – 09:25	15 min	**Part 1: Motivation & Accelerator Physics Foundations** Operational stability requirements, Hill's equation, betatron oscillation, ORM derivation, and hackathon lattice geometry.
09:25 – 09:45	20 min	**Part 2: Supervised Learning & Neural Surrogate Modeling** Simulation tracking bottlenecks, offline exploration, Linear Ridge vs. Multi-Layer Perceptron (MLP), modeling cubic core saturation.
09:45 – 10:10	25 min	**Part 3: Closed-Loop Control Paradigms** Baseline 0 (Classical SVD pseudo-inverse), Baseline 1 (SciPy `L-BFGS-B` digital twin inversion), and Track 2 (RL policy agent with dynamic damping).
10:10 – 10:25	15 min	**Part 4: Mini-Hackathon Problem Formulation & Gymnasium Contract** Environment spaces, reward formulation, 30-seed evaluation protocol, scoring formula trade-offs, and live Weights & Biases (W&B) telemetry.
10:25 – 10:30	5 min	**Part 5: Summary, Q&A & Transition to Lecture 2** Key takeaways and transition to Dr. Thorsten Hellert's Agentic AI & Osprey session.

Table 1: Curriculum timeline for the 80-minute morning session.

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2 Part 1: Motivation & Accelerator Physics Foundations (15 min)

2.1 The Operational Stability Challenge

In synchrotron light sources, free-electron lasers (XFELs), and medical linacs, particle beam stability must be sustained at sub-micron levels to maintain photon brightness and collision luminosity. Orbit distortions arise from multiple sources:

- **Slow Drifts:** Thermal expansion of girders, diurnal ambient temperature shifts, and ground motion.
- **Dynamic Jitter:** Mechanical vibration from cooling water pumps, cryocoolers, and power supply ripple.
- **High Actuator Coupling:** Deflecting a single dipole corrector causes betatron wave propagation across the entire transport line.

2.2 Transverse Beam Dynamics & Hill's Equation

In an uncoupled linear transport channel, the transverse displacement $x(s)$ as a function of path length s is governed by **Hill's Equation**:

$$\frac{d^2x(s)}{ds^2} + K(s)x(s) = 0 \tag{1}$$

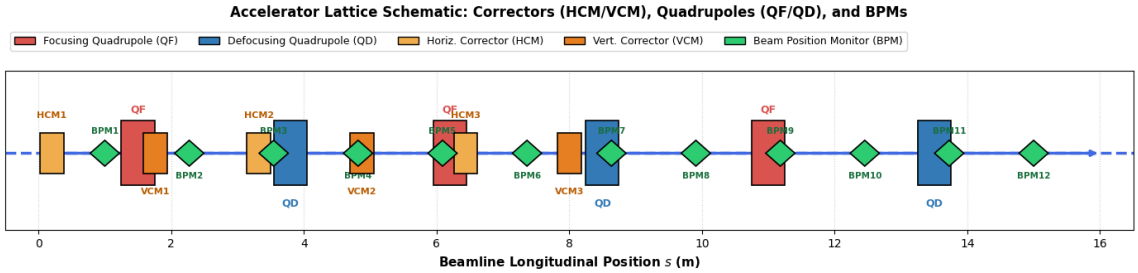


Figure 1: Schematic diagram of the 15-meter accelerator transport section showing alternating focusing (QF, red) and defocusing (QD, blue) quadrupoles, 6 dipole steerers (HCM/VCM), and 12 Beam Position Monitors.

where $K(s) = \frac{e}{p}\frac{\partial B_y(s)}{\partial x}$ represents the alternating quadrupole focusing strength ($K > 0$ for horizontal focusing 'QF', $K < 0$ for defocusing 'QD').

The closed-form betatron solution is given by:

$$x(s) = \sqrt{\epsilon\beta(s)}\cos\left(\psi(s) + \phi_0\right) \tag{2}$$

where $\beta(s)$ is the Courant-Snyder beta function, $\psi(s) = \int_0^s \frac{d\bar{s}}{\beta(\bar{s})}$ is the betatron phase advance, and $\nu = \frac{1}{2\pi}\oint \frac{ds}{\beta(s)}$ is the betatron tune.

2.3 Derivation of the Orbit Response Matrix (ORM)

When an angular kick $\theta_j = \Delta x'_j = \frac{eBL}{p}$ is applied by a dipole corrector at longitudinal position s_j, the transverse displacement registered at a downstream Beam Position Monitor (BPM) at position s_i is linearly proportional to the kick angle:

$$R_{ij} = \frac{\Delta x(s_i)}{\theta_j} = \frac{\sqrt{\beta(s_i)\beta(s_j)}}{2\sin(\pi\nu)}\cos\left(|\psi(s_i) - \psi(s_j)| - \pi\nu\right) \tag{3}$$

For M BPMs and N dipole steerers, the linear response is encapsulated by the matrix $\mathbf{R} \in \mathbb{R}^{N\times M}$:

$$\mathbf{Y}_{\text{linear}} = \mathbf{I}\cdot\mathbf{R} \tag{4}$$

where $\mathbf{I} = [I_1, I_2, \ldots, I_N]$ represents the corrector current vector.

2.4 Mini-Hackathon Lattice Geometry & Realistic Non-Linearities

The mini-hackathon benchmark lattice (Figure 1) models a 15-meter transport line:

- **Actuators:** 6 dipole steerers (3 Horizontal 'HCM1–3', 3 Vertical 'VCM1–3') bounded in $[-5.0\text{ A}, +5.0\text{ A}]$, with single-step adjustments clipped to $[-0.8\text{ A}, +0.8\text{ A}]$.
- **Diagnostics:** 12 Beam Position Monitors ($s = 1.0$ m to 15.0 m) reading beam displacement in millimeters.
- **Magnetic Core Saturation:** Ferromagnetic saturation at higher excitation currents induces non-linear cubic rolloff:

$$\mathbf{Y}_{\text{sat}} = -0.015\cdot(\mathbf{I}^3\cdot\mathbf{R}) \tag{5}$$

- **Initial Misalignment Distortion ($\mathbf{Y}_0$):** Quadrupole displacement errors drawn per episode:

$$Y_{0,k} = A_1\sin\left(\frac{2\pi s_k}{\lambda_1} + \phi_1\right) + A_2\cos\left(\frac{2\pi s_k}{\lambda_2} + \phi_2\right) \tag{6}$$

  producing initial orbit errors between 0.4 mm and 1.1 mm RMS.

- **Gaussian Measurement Noise:** Sensor electronics introduce random noise $\boldsymbol{\epsilon} \sim \mathcal{N}(0, \sigma^2)$, with $\sigma = 0.010$ mm (10 μm).

The complete physical forward response equation is:

$$\mathbf{Y} = \mathbf{I} \cdot \mathbf{R} - 0.015 \cdot (\mathbf{I}^3 \cdot \mathbf{R}) + \mathbf{Y}_0 + \boldsymbol{\epsilon} \tag{7}$$

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3 Part 2: Supervised Learning & Neural Surrogate Modeling (20 min)

3.1 Why Surrogate Modeling?

Numerical tracking engines (e.g., Elegant, MAD-X, OPAL) accurately simulate collective effects and 3D space charge, but require significant computational time. In contrast, closed-loop beam feedback requires microsecond-level response times. AI **Surrogate Models (Digital Twins)** learn the input-output mapping offline, enabling ultra-fast forward evaluations.

3.2 Dataset Generation & Validation Partitioning

In Step 3 of the challenge notebook, $1,500$ experimental samples are synthesized uniformly across current boundaries $[-3.0\text{ A}, +3.0\text{ A}]$:

```
n_train_samples = 1500
X_offline = np.random.uniform(-3.0, 3.0, size=(n_train_samples, env.
    n_correctors))
Y_linear = X_offline @ env.R
Y_sat = - 0.015 * (X_offline**3) @ env.R
bpm_noise = np.random.normal(0, 0.010, size=(n_train_samples, env.n_bpms))
Y_offline = Y_linear + Y_sat + bpm_noise

X_train, X_test, Y_train, Y_test = train_test_split(
    X_offline, Y_offline, test_size=0.20, random_state=42
)
```

3.3 Model Architecture Comparison: Ridge vs. Multi-Layer Perceptron

We compare two supervised regression architectures:

1. **Regularized Linear Ridge Regression:**

$$\min_{\mathbf{W}} \|\mathbf{Y} - \mathbf{X}\mathbf{W}\|_F^2 + \lambda \|\mathbf{W}\|_F^2 \tag{8}$$

   Because Ridge is strictly linear, it fails to account for the cubic saturation term ($\mathbf{I}^3$).

2. **Deep Multi-Layer Perceptron (MLP):**

$$\hat{\mathbf{Y}} = f_{\boldsymbol{\theta}}(\mathbf{I}) = \mathbf{W}_3 \cdot \mathrm{ReLU}\left(\mathbf{W}_2 \cdot \mathrm{ReLU}\left(\mathbf{W}_1 \mathbf{I} + \mathbf{b}_1\right) + \mathbf{b}_2\right) + \mathbf{b}_3 \tag{9}$$

   Constructed with hidden layers $(128, 64, 32)$ and trained via the Adam optimizer (L_2 weight penalty $\alpha = 10^{-4}$).

Model Architecture	Parameters	Holdout RMSE	Inference Latency	Physical Regimes Captured
Linear Ridge Baseline ($\lambda = 1.0$)	72	0.0558 mm	< 0.01 ms	Linear optics only; misses saturation.
MLP Neural Surrogate	11,596	**0.0118** mm	≈ 0.15 ms	Linear ORM and cubic saturation down to noise floor.

Table 2: Holdout validation performance on 20% test partition (300 unseen shots).

Physical Insight: The Bayesian Sensor Noise Floor

The holdout error of the MLP (0.0118 mm) closely approaches the theoretical noise floor set by Gaussian BPM sensor jitter ($\sigma = 0.0100$ mm). This indicates that the neural surrogate has learned the non-linear physics without overfitting sensor noise.

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4 Part 3: Closed-Loop Control Paradigms (25 min)

4.1 Baseline 0: Classical SVD Pseudo-Inverse Controller

In standard accelerator control rooms, Singular Value Decomposition (SVD) of the transpose response matrix $\mathbf{R}^T = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T$ provides the Moore-Penrose pseudo-inverse:

$$(\mathbf{R}^T)^\dagger = \mathbf{V}\mathbf{\Sigma}^{-1}\mathbf{U}^T \tag{10}$$

The closed-loop correction action per step is:

$$\Delta\mathbf{I} = \text{clip}\left(-\alpha \cdot (\mathbf{R}^T)^\dagger \cdot \mathbf{Y}_{\text{obs}},\ -0.8,\ +0.8\right) \tag{11}$$

where $\alpha \approx 0.45$ is the feedback gain. While computationally instantaneous, SVD assumes linearity; when correctors saturate, high gains trigger overshooting and limit-cycle oscillations.

4.2 Baseline 1: Model-Based Optimization via Digital Twin Inversion

Rather than inverting an analytical matrix, Baseline 1 optimizes corrective kicks directly through the forward neural surrogate $f_{\boldsymbol{\theta}}(\Delta\mathbf{I})$ using SciPy's `L-BFGS-B` algorithm:

$$\min_{\Delta\mathbf{I}} \mathcal{L}(\Delta\mathbf{I}) = \|\mathbf{Y}_{\text{obs}} + f_{\boldsymbol{\theta}}(\Delta\mathbf{I})\|_2^2 + \lambda\|\Delta\mathbf{I}\|_2^2 \quad \text{s.t.} \quad \Delta\mathbf{I} \in [-5.0\text{ A}, +5.0\text{ A}]^6 \tag{12}$$

```python
class SupervisedMLPController:
    def __init__(self, surrogate):
        self.surrogate = surrogate

    def act(self, obs, info=None):
        def objective(delta_I):
            pred = self.surrogate.predict(delta_I.reshape(1, -1))[0]
            residual = obs + pred
            return np.sum(residual**2) + 0.0001 * np.sum(delta_I**2)

        res = opt.minimize(objective, np.zeros(6), bounds=[(-5.0, 5.0)]*6,
    method='L-BFGS-B')
        return np.clip(res.x, -0.8, 0.8)
```

Because the surrogate anticipates non-linear saturation falloff, Baseline 1 reaches convergence in only **2 to 3 iterations**.

4.3 Track 2: Closed-Loop RL Policy with Physics Prior and Adaptive Damping

Model-based iterative optimization requires 15–30 ms per step. A closed-loop policy network executes in < 0.5 ms. High-performing accelerator policy agents integrate an analytical **physics prior** with a learned parameter residual and dynamic step damping:

$$\Delta\mathbf{I}(t) = \gamma(t) \cdot \left[\underbrace{-\alpha(\mathbf{R}^T)^{\dagger}\mathbf{Y}}_{\text{Analytical Physics Prior}} + \underbrace{\mathbf{W}_{\text{learned}}\mathbf{Y} + \mathbf{b}}_{\text{Learned Residual Offset}} \right] \tag{13}$$

$$\gamma(t) = \frac{1.0}{1.0 + 0.03 \cdot t} \tag{14}$$

The dynamic damping factor $\gamma(t)$ decays control authority as the beam approaches the 0.030 mm boundary, preventing sensor noise amplification from destabilizing the closed orbit.

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5 Part 4: Mini-Hackathon Problem Formulation & Gymnasium Contract (15 min)

5.1 Gymnasium Environment Specification (`AcceleratorOrbitGymEnv`)

- **Observation Space:** $\text{Box}(-10.0, +10.0, \text{shape} = (12,), \text{dtype} = \text{float32})$ (BPM readings in mm).
- **Action Space:** $\text{Box}(-0.8, +0.8, \text{shape} = (6,), \text{dtype} = \text{float32})$ (Incremental kicks $\Delta\mathbf{I}$ in Amperes).
- **State Update:** $\mathbf{I}_{t+1} = \text{clip}(\mathbf{I}_t + \Delta\mathbf{I}_t, -5.0, 5.0)$.
- **Reward Function:**

$$r_t = -\text{RMS}_{t+1} - 0.01 \cdot \|\Delta\mathbf{I}_t\|_2 + (50.0 \text{ if } \text{RMS}_{t+1} < 0.030 \text{ mm else } 0.0) \tag{15}$$

- **Success Criteria:** Terminated when RMS < 0.030 mm; truncated if step count reaches 20.

5.2 Standardized 30-Seed Evaluation & Official Scoring Formula

Controllers are evaluated across 30 standardized seeds (`range(1, 31)`). The official benchmark score is defined as:

$$\text{Score} = (\text{Success Rate} \times 10.0) - (2.0 \times \overline{\text{Steps}}) + \left(\frac{0.03}{\max(\overline{\text{RMS}}, 0.005)} \times 20.0 \right) - (2.0 \times \overline{\|\mathbf{I}\|}) \tag{16}$$

Rank	Controller Paradigm	Success Rate	Mean Steps	Final RMS	Current Norm	Official Score
#1	Baseline 1: Supervised MLP Inversion	100.0%	**2.3 steps**	0.0257 mm	1.21 A	**1,016.8 pts**
#2	Baseline 0: Classical Linear SVD	100.0%	5.6 steps	**0.0230 mm**	**1.16 A**	**1,013.1 pts**
#3	Track 2: Closed-Loop RL Policy	100.0%	6.1 steps	0.0239 mm	1.18 A	**1,011.1 pts**

Table 3: Official benchmark results across 30 evaluation seeds.

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6 Part 5: Summary, Q&A & Transition to Lecture 2 (5 min)

Summary Takeaways

1. **Physics-Informed Prior:** Initializing AI policies with the analytical pseudo-inverse ORM guarantees early stability and avoids millions of exploratory random steps.
2. **Non-Linear Representation:** Deep neural surrogates bridge the gap between idealized linear beam optics and real-world magnetic saturation.
3. **Dynamic Step Damping:** Attenuating feedback near diagnostic noise floors prevents terminal limit cycles and ensures robust convergence.

Next Session: Dr. Thorsten Hellert will present *Agentic AI for Accelerator Research: Osprey Applications* (10:40 – 12:00), exploring LLM agents, automated tool calling, and autonomous accelerator orchestration.

References:

1. Minty, M. G., & Zimmermann, F. (2003). *Measurement and Control of Charged Particle Beams*. Springer.
2. Wiedemann, H. (2015). *Particle Accelerator Physics* (4th ed.). Springer.
3. Hellert, T., et al. (2024). *Autonomous Accelerator Tuning using Large Language Model Agents and Osprey*. Phys. Rev. Accel. Beams.
4. Edelen, A., et al. (2020). *Machine learning for orders-of-magnitude speedup in multiobjective optimization of particle accelerator beamlines*. Phys. Rev. Accel. Beams, 23(4), 044601.